11. Write the python program for Map Coloring to implement CSP.

Aim: To implement the Map Coloring Problem using Constraint Satisfaction Problem (CSP).

Procedure:

1. Define the regions and their adjacent regions.
2. Define the available colors.
3. Select a region for coloring.
4. Check whether the selected color conflicts with neighboring regions.
5. Use backtracking to try another color if a conflict occurs.
6. Display the valid color assignment.

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```
regions = ['A', 'B', 'C', 'D']
colors = ['Red', 'Green', 'Blue']

graph = {
    'A': ['B', 'C'],
    'B': ['A', 'C', 'D'],
    'C': ['A', 'B', 'D'],
    'D': ['B', 'C']
}
coloring = {}

def solve(index):
    if index == len(regions):
        return True

    region = regions[index]

    for color in colors:
        if all(coloring.get(n) != color
               for n in graph[region]):
            coloring[region] = color

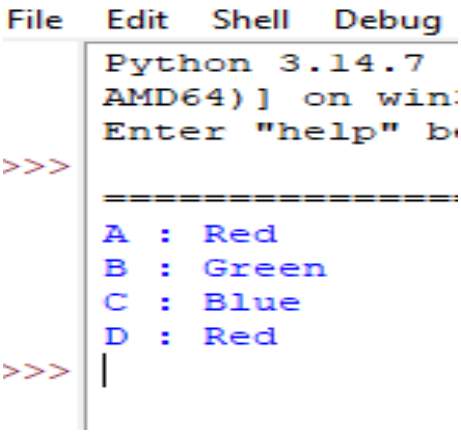
            if solve(index + 1):
                return True

            del coloring[region]

    return False

if solve(0):
    for region, color in coloring.items():
        print(region, ":", color)
else:
    print("No solution")
```

Output:



Result: Thus, the Map Coloring Problem was implemented successfully using CSP.